ProofPass

Privacy-Preserving Digital Identity and Credential Verification on BOT Chain

Abstract

As digital interactions become ubiquitous, the necessity for robust, verifiable identity systems has grown exponentially. Current centralized paradigms force individuals to relinquish control over sensitive personal data, leading to massive breaches and loss of privacy. ProofPass introduces a decentralized, privacy-preserving credential verification architecture deployed natively on BOT Chain. Utilizing cryptographic commitments and zero-knowledge hash-chain proofs, ProofPass enables the issuance, selective disclosure, and cryptographic verification of credentials—ranging from academic degrees to age eligibility—without exposing the underlying raw data on any public ledger.

1. Introduction

The cornerstone of modern digital trust is the ability to prove claims about oneself. However, traditional credentialing is severely flawed. Users are routinely required to photograph government-issued IDs or share comprehensive academic transcripts simply to prove a binary claim, such as "I am over 18 years of age."

ProofPass solves this by separating the credential payload from its onchain verification state. By utilizing BOT Chain for immutable state management (issuer registries and revocation lists) and offchain data sharing coupled with genuine cryptographic proofs, ProofPass achieves a trustless, scalable, and fully private digital identity ecosystem.

2. Problem Statement

- **Over-disclosure of Data:** Users are forced to share excess personal information (e.g., date of birth, home address) for simple verifications.
- **Centralized Honeypots:** Traditional databases storing identity records are prime targets for cyberattacks.
- **Lack of Immutability and Auditability:** Paper credentials can be forged, and centralized digital records can be silently altered or revoked without a transparent audit trail.

3. Proposed Solution & Architecture

ProofPass operates on a triad model consisting of three distinct roles, all interacting through the high-throughput, secure infrastructure of BOT Chain:

Role	Responsibility	BOT Chain Interaction
Issuer	Universities, Governments, Employers	Registers DID, commits credential hashes, manages revocation states.
Holder	Individual End-Users	Stores raw data locally. Does NOT write to chain directly. Generates proofs.
Verifier	Services, Venues, Auditors	Reads chain state to verify commitments and check revocation lists.

Table 1: The ProofPass Triad Architecture

4. Credential Issuance & Cryptographic Commitments

To preserve privacy, a credential's raw data is never submitted to a smart contract. Instead, the Issuer generates a cryptographic commitment. For a standard credential, this is computed as:

```
C = SHA256( Holder_Address + Salt + JSON(Credential_Data) )
```

The value `C` is written to the `ProofPassRegistry` smart contract. Because SHA-256 is collision-resistant and pre-image resistant, observing `C` on BOT Chain provides zero information about the Holder's identity or the credential contents.

5. Zero-Knowledge Technology

A core feature of ProofPass is its ability to support genuine Zero-Knowledge Range Proofs, specifically tailored for age eligibility or numerical thresholds, without relying on heavy zk-SNARK circuits that are difficult to process in pure frontend environments. ProofPass achieves this via *Cryptographic Hash Chains*.

The Hash Chain Algorithm:
Let `S` be a high-entropy secure seed. The Issuer applies a cryptographic hash function `N` times (where `N` is the maximum possible age, e.g., 100). The public commitment is $H^{100}(S)$.

If a Holder is `A` years old, the Issuer provides them with the secret $H^{100-A}(S)$. To prove to a Verifier that they meet a threshold age `T` (where $A \geq T$), the Holder computes and reveals:

$$Proof = H^{A-T}(H^{100-A}(S)) = H^{100-T}(S)$$

The Verifier then hashes this *Proof* exactly `T` times. If the result equals the onchain commitment $H^{100}(S)$, the proof is mathematically valid. Due to the one-way nature of the hash function, the Verifier cannot deduce the exact age `A`, thus achieving zero-knowledge threshold verification.

6. Revocation System

Credentials may need to be revoked (e.g., an expired membership or a rescinded degree). The BOT Chain `ProofPassRegistry` includes a mapping: `mapping(bytes32 ⇒ bool) public isRevoked;`.

By executing `revokeCredential(bytes32 commitment)`, an Issuer flips the boolean flag to `true`. When a Verifier checks the proof, they must independently query the contract's `isValid(commitment)` function, which returns false if the credential has been revoked, ensuring real-time validity checks without relying on centralized APIs.

7. Smart Contract Architecture

The primary Solidity smart contract handles three critical functions with minimal gas footprint:

1. **Issuer Authorization:** Only whitelisted addresses (governed by the protocol Admin or a DAO) can commit credentials, preventing spam and establishing trust roots.
2. **Commitment Storage:** Efficiently storing 32-byte hashes linked to the Issuer's address.
3. **State Queries:** Fast `view` functions allowing Verifiers to confirm `credentialToIssuer[C] ≠ address(0)` and `isRevoked[C] == false`.

Because no personal data touches the Ethereum Virtual Machine (EVM) state, the contracts are fully compliant with global data protection regulations (such as GDPR), as no "right to be forgotten" conflicts occur on the immutable ledger.

8. Use Cases

- **Student Identity:** Proving university enrollment for software discounts without revealing student ID numbers or academic transcripts.
- **Age-Restricted Commerce:** Purchasing restricted goods online by submitting a zero-knowledge hash chain proof proving age ≥ 18 or 21.
- **Employment Verification:** Allowing prospective employers to verify past employment instantly against cryptographic commitments issued by previous HR departments.

9. Roadmap & Conclusion

ProofPass represents a paradigm shift in how identity is managed. By leveraging BOT Chain's high-speed consensus and robust smart contract environment, coupled with elegant cryptographic primitives like hash chains and commitment schemes, ProofPass delivers a decentralized, highly private credentialing network.

Future phases will include the integration of fully-fledged zk-SNARK circuits (via WASM implementations) to allow for complex multi-claim arbitrary logic proofs, further cementing ProofPass as the premier identity solution on BOT Chain.